(23.09) Física Electrónica - Hojas de fórmulas

Cristales semiconductores

$$f(E) = \frac{1}{1+e^{(E-E_F)/kT}}$$

$$g_C(E) = \frac{m_n^* \sqrt{2m_n^*(E-E_C)}}{\pi^2\hbar^3} \quad , \quad E \geq E_C$$

$$g_V(E) = \frac{m_p^* \sqrt{2m_p^*(E_V-E)}}{\pi^2\hbar^3} \quad , \quad E \leq E_V$$

$$n = n_i \cdot e^{(E_F-E_i)/kT}$$
$$p = n_i \cdot e^{(E_i-E_F)/kT}$$

$$n \cdot p = n_i^2$$
$$p - n + N_D - N_A = 0$$

$$n = \frac{N_D-N_A}{2} + \sqrt{\left(\frac{N_D-N_A}{2}\right)^2 + n_i^2}$$
$$p = \frac{N_A-N_D}{2} + \sqrt{\left(\frac{N_A-N_D}{2}\right)^2 + n_i^2}$$

$$E_i = \frac{E_C+E_V}{2} + \frac{3}{4}kT \cdot \ln\left(\frac{m_p^*}{m_n^*}\right)$$
$$E_F - E_i = kT \cdot \ln\left(\frac{N_D}{n_i}\right)$$
$$E_i - E_F = kT \cdot \ln\left(\frac{N_A}{n_i}\right)$$

Corrientes

$$\vec{J} = q \cdot (n \cdot \vec{v_n} + p \cdot \vec{v_p})$$
$$\vec{J} = \sigma \cdot \vec{E}$$
$$\sigma = q \cdot (\mu_n \cdot n + \mu_p \cdot p)$$
$$\rho = 1/\sigma$$
$$\vec{v} = \frac{\mu_0 \cdot \vec{E}}{\sqrt[\beta]{1+\left(\frac{\mu_0 E}{v_{sat}}\right)^\beta}} = \mu \cdot \vec{E}$$

$$\vec{J} = q \cdot (D_N \vec{\nabla} n - D_P \vec{\nabla} p)$$

$$D_P = \mu_P \cdot \frac{kT}{q} \ , \ D_N = \mu_N \cdot \frac{kT}{q}$$

$$L_P = \sqrt{D_P \cdot \tau_P} \ , \ L_N = \sqrt{D_N \cdot \tau_N}$$

$$\vec{J} = \vec{J_P} + \vec{J_N}$$

$$\vec{J_P} = \vec{J}_{P|\text{Drift}} + \vec{J}_{P|\text{Diff}}$$

$$\vec{J_N} = \vec{J}_{N|\text{Drift}} + \vec{J}_{N|\text{Diff}}$$

Efecto Hall

$$v_H = \frac{I}{qNt} \cdot B = V \cdot \mu \frac{W}{L} \cdot B$$

$$R = \rho \cdot \frac{L}{A} = \frac{L}{Aq\mu N} \quad , \quad A = Wt$$

$$S_H^I = \frac{1}{qNt} \quad , \quad S_H^V = \mu \frac{W}{L}$$

Magnitudes físicas

$$\vec{E}(x,y,z,t) = -\vec{\nabla} V(x,y,z,t)$$

$$\rho(x,y,z,t) = \epsilon_r \epsilon_0 \left(\vec{\nabla} \cdot \vec{E}(x,y,z,t)\right)$$

$$\rho(x,y,z,t) = -\epsilon_r \epsilon_0 \nabla^2 V(x,y,z,t)$$

$$I = \iint_S \vec{J}(x,y,z,t) \cdot d\vec{S}$$

Juntura P-N (1-D, abrupta)

$$V_{bi} = V_T \cdot \ln\left(\frac{N_A \cdot N_D}{n_i^2}\right) \quad , \quad V_T = \frac{kT}{q} \approx 26mV \ (T = 300K)$$

$$x_n(V) = \sqrt{\frac{2\epsilon_r\epsilon_0}{q} \cdot \frac{N_A}{N_D \cdot (N_A+N_D)} \cdot (V_{bi} - V)}$$

$$x_p(V) = \sqrt{\frac{2\epsilon_r\epsilon_0}{q} \cdot \frac{N_D}{N_A \cdot (N_A+N_D)} \cdot (V_{bi} - V)}$$

$$W(V) = x_n(V) + x_p(V) = \sqrt{\frac{2\epsilon_r\epsilon_0}{q} \cdot \frac{N_A+N_D}{N_A \cdot N_D} \cdot (V_{bi} - V)}$$

$$V(x) = \begin{cases} 0 & , x < -x_p \\ \frac{qN_A}{2\epsilon_r\epsilon_0} \cdot (x_p + x)^2 & , -x_p \leq x \leq 0 \\ V_{bi} - \frac{qN_D}{2\epsilon_r\epsilon_0} \cdot (x_n - x)^2 & , 0 < x \leq x_n \\ V_{bi} & , x > x_n \end{cases}$$

$$E_x(x) = \begin{cases} -\frac{qN_A}{\epsilon_r\epsilon_0}(x_p + x) & , -x_p \leq x \leq 0 \\ -\frac{qN_D}{\epsilon_r\epsilon_0}(x_n - x) & , 0 < x \leq x_n \\ 0 & , x < -x_p \vee x > x_n \end{cases}$$

$$\rho(x) = \begin{cases} -qN_A & , -x_p \leq x \leq 0 \\ qN_D & , 0 < x \leq x_n \\ 0 & , x < -x_p \vee x > x_n \end{cases}$$

$$N_A \cdot x_p = N_D \cdot x_n$$

$$\gamma_P = \frac{J_P}{J_P + J_N}$$

Juntura M-S (Tipo n, $\phi_M > \phi_S$)

$$\Phi_B = \Phi_M - \chi$$

$$V_{bi} = \frac{\Phi_B - (E_C - E_F)_{FB}}{q}$$

$$(E_C - E_F)_{FB} \approx \frac{E_G}{2} - (E_F - E_i) = \frac{E_G}{2} - kT \cdot ln\left(\frac{N_D}{n_i}\right)$$

$$\rho(x) = \begin{cases} -qN_D\delta & , x = 0 \\ qN_D & , 0 < x \leq W \\ 0 & , x > W \vee x < 0 \end{cases}$$

$$E_x(x) = \begin{cases} \frac{-qN_D}{\epsilon_r\epsilon_0}(W - x) & , 0 \leq x \leq W \\ 0 & , x > W \end{cases}$$

$$V(x) = \begin{cases} \frac{-qN_D}{2\epsilon_r\epsilon_0}(W - x)^2 & , 0 \leq x \leq W \\ 0 & , x > W \end{cases}$$

$$W(V) = \sqrt{\frac{2\epsilon_r\epsilon_0}{qN_D}(V_{bi} - V)}$$

Diodo rectificador

$$I_D(V) = I_S \cdot (e^{V/V_T} - 1)$$

$$I_S = qA \cdot \left(\frac{D_N}{L_N} \cdot \frac{n_i^2}{N_A} + \frac{D_P}{L_P} \cdot \frac{n_i^2}{N_D}\right)$$

$$I_S = A \cdot \mathcal{A} \cdot T^2 \cdot e^{-\phi_B/kT} \quad , \quad \mathcal{A} = \frac{4\pi m_e k^2 q}{h^3}$$

$$r_e = \frac{V}{I_D(V)} = \frac{V}{I_S \cdot (e^{V/V_T} - 1)}$$

$$r_d = \frac{\partial V}{\partial I_D(V)} = \frac{1}{\frac{\partial I_D(V)}{\partial V}} = \frac{V_T}{(I_D(V) + I_S)} = \frac{V_T}{I_S} \cdot e^{-V/V_T}$$

$$C'_j = \frac{C_j}{A} = \frac{\epsilon_r\epsilon_0}{W(V)} = \frac{C'_{j0}}{\sqrt{1 - \frac{V}{V_{bi}}}} \ ; Cj_0 = C_j(V = 0)$$

$$C'_D = \frac{C_D}{A} = \frac{q}{V_T}\left(L_n n_{p0} + L_p p_{n0}\right) e^{V/V_T}$$

Colaboraciones:
Kammann, Lucas Agustín
Farall, Facundo David
Mechoulam, Alan
Lambertucci, Guido
Müller, Malena
Smolkin, Pablo
Pedrosa, Carola
Dutriez, Philippe

BJT (PNP)

$$\Delta p_B(x) = \Delta p_B(0)\left(1-\tfrac{x}{W}\right) + \Delta p_B(W)\,\tfrac{x}{W}$$

$$\Delta n_E(0) = n_{E0}\left(e^{V_{EB}/V_T}-1\right) \quad , \quad \Delta n_E(+\infty) = 0$$

$$\Delta p_B(0) = p_{B0}\left(e^{V_{EB}/V_T}-1\right)$$

$$\Delta p_B(W) = p_{B0}\left(e^{V_{CB}/V_T}-1\right)$$

$$\Delta n_C(0) = n_{C0}\left(e^{V_{CB}/V_T}-1\right) \quad , \quad \Delta n_C(+\infty) = 0$$

$$I_{E_p} = q\cdot A\cdot \tfrac{D_B}{L_B}\cdot p_{B0}\left(\frac{e^{\frac{V_{EB}}{V_T}}-1}{\tanh\left(\frac{W}{L_B}\right)} - \frac{e^{\frac{V_{CB}}{V_T}}-1}{\sinh\left(\frac{W}{L_B}\right)}\right)$$

$$I_{C_p} = q\cdot A\cdot \tfrac{D_B}{L_B}\cdot p_{B0}\left(\frac{e^{\frac{V_{EB}}{V_T}}-1}{\sinh\left(\frac{W}{L_B}\right)} - \frac{e^{\frac{V_{CB}}{V_T}}-1}{\tanh\left(\frac{W}{L_B}\right)}\right)$$

$$I_{E_n} = q\cdot A\cdot \tfrac{D_E}{L_E}\cdot n_{E0}\left(e^{\frac{V_{EB}}{V_T}}-1\right)$$

$$I_{C_n} = -q\cdot A\cdot \tfrac{D_C}{L_C}\cdot n_{C0}\left(e^{\frac{V_{CB}}{V_T}}-1\right)$$

$$I_{RB} = I_{E_p} - I_{C_p} \quad , \quad I_B = I_{E_n} + I_{C_n} + I_{RB}$$

$$\alpha_T = \tfrac{IC_p}{IE_p} \approx \tfrac{1}{1+\frac{1}{2}\cdot\left(\frac{W}{L_B}\right)^2} \quad , \quad = \tfrac{I_{E_p}}{I_E} \approx \tfrac{1}{1+\frac{D_E}{D_B}\cdot\frac{N_B}{N_E}\cdot\frac{W}{L_E}}$$

$$\alpha = \gamma\cdot\alpha_T \approx \tfrac{1}{1+\frac{D_E}{D_B}\cdot\frac{N_B}{N_E}\cdot\frac{W}{L_E}+\frac{1}{2}\cdot\left(\frac{W}{L_B}\right)^2}$$

$$\beta = \tfrac{I_C}{I_B} = \tfrac{\alpha}{1-\alpha} \approx \tfrac{1}{\frac{D_E}{D_B}\cdot\frac{N_B}{N_E}\cdot\frac{W}{L_E}+\frac{1}{2}\cdot\left(\frac{W}{L_B}\right)^2}$$

$$I_E(V_{EB},V_{CB}) = I_{F0}\cdot\left(e^{\frac{V_{EB}}{V_T}}-1\right) - \alpha_R\cdot I_{R0}\left(e^{\frac{V_{CB}}{V_T}}-1\right)$$

$$I_C(V_{EB},V_{CB}) = \alpha_F\cdot I_{F0}\cdot\left(e^{\frac{V_{EB}}{V_T}}-1\right) - I_{R0}\cdot\left(e^{\frac{V_{CB}}{V_T}}-1\right)$$

$$I_B(V_{EB},V_{CB}) = I_E(V_{EB},V_{CB}) - I_C(V_{EB},V_{CB})$$

$$I_C = \alpha\cdot I_E + I_{CB0} \quad , \quad I_C = \beta\cdot I_B + I_{CE0}$$

$$r_\pi = \tfrac{\beta}{g_m} \quad , \quad g_m = \tfrac{I_C}{V_T} \quad , \quad r_o = \tfrac{V_A}{I_C}$$

$$r_\mu \approx \beta_o\cdot r_o \quad , \quad g_m\cdot r_o = \tfrac{V_A}{V_T}$$

$$C_\pi(I_C) = C_b(I_C) + C_{je} \quad , \quad C_b(I_C) = \tau_B\cdot g_m(I_C)$$

$$C_{je} \approx 2C_{je0} \quad , \quad C_\mu(V_{BC}) = \tfrac{C_{\mu 0}}{(1-\frac{V_{BC}}{V_{bi}})^{n_c}} \; , \; f_T = \tfrac{1}{2\pi}\tfrac{g_m}{C_\pi + C_\mu}$$

J-FET (canal n)

$$V_P = V_{bi} - \tfrac{qN_Ca^2}{2\epsilon_r\epsilon_0} \quad , \quad a = \sqrt{\tfrac{2\epsilon_r\epsilon_0}{qN_C}(V_{bi}-V_P)}$$

$$I_{DSS} = \tfrac{G_0\cdot V_P}{2} \quad , \quad G_0 = q\mu_n N_C\tfrac{2aZ}{L}$$

$$I_{Dsat} = \tfrac{G_0\cdot V_{Dsat}^2}{2V_P} \quad , \quad V_{Dsat} = V_{GS} - V_P$$

$$I_D(V_{GS},V_{DS}) = \begin{cases} \frac{G_0}{V_P}\left[(V_{GS}-V_P)\cdot V_{DS} - \frac{V_{DS}^2}{2}\right], 0 \le V_{DS} \le V_{D_{sat}} \wedge V_P \le V_{GS} \\ I_{DSS}\cdot\left(1-\frac{V_{GS}}{V_P}\right)^2 \qquad\qquad , V_{DS} > V_{D_{sat}} \wedge V_P \le V_{GS} \end{cases}$$

$$g_m = -2\cdot\tfrac{I_{DSS}}{V_P}\cdot(1-\tfrac{V_{GS}}{V_P}) \quad , \quad r_o = \tfrac{V_A}{I_D}$$

Transistor MOSFET (Enriquecimiento, canal n)

$$\phi_F = V_T\ln\left(\tfrac{N_A}{n_i}\right) \quad , \quad W = \sqrt{\tfrac{2\epsilon_r\epsilon_0}{qN_A}\cdot\phi_s} \quad , \quad W_T = \sqrt{\tfrac{2\epsilon_{rs}\epsilon_0}{qN_A}\cdot 2\phi_F}$$

$$C'_{ox} = \tfrac{C_{ox}}{A} = \tfrac{\epsilon_{rox}\epsilon_0}{t_{ox}} \quad , \quad C'_S = \tfrac{C_S}{A} = \tfrac{\epsilon_{rs}\epsilon_0}{W}$$

$$C_{MOS} = \begin{cases} C_{ox} & , \; Acc \\ \frac{C_{ox}C_S(W)}{C_{ox}+C_S(W)} & , \; Dep \\ C_{ox} & , \; Inv\,(\omega\to 0) \\ \frac{C_{ox}C_S(W_T)}{C_{ox}+C_S(W_T)} & , \; Inv\,(\omega\to\infty) \end{cases}$$

$$V_{TH0} = 2\phi_F + \gamma\cdot\sqrt{2\phi_F} \quad , \quad \gamma = \tfrac{1}{C'_{ox}}\sqrt{2q\epsilon_{rs}\epsilon_0 N_A} = \tfrac{t_{ox}}{\epsilon_{rox}}\sqrt{\tfrac{2qN_A\epsilon_{rs}}{\epsilon_0}}$$

$$V_{TH}(V_{SB}) = V_{TH0} + \gamma\cdot(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}) \quad , \quad V_{Dsat} = V_{GS} - V_{TH}$$

$$I_D(V_{GS},V_{DS}) = \begin{cases} I_D = \mu C'_{ox}\frac{W}{L}\left((V_{GS}-V_{TH})V_{DS} - \frac{V_{DS}^2}{2}\right), V_{DS} \le V_{Dsat} \\ I_D = \frac{\mu C'_{ox}}{2}\frac{W}{L}(V_{GS}-V_{TH})^2, V_{DS} > V_{Dsat} \end{cases}$$

$$I_D(V_A) = I_D\cdot(1+\tfrac{V_{DS}}{V_A}) = I_D\cdot(1+\lambda\cdot V_{DS})$$

$$g_m = \mu_n C'_{ox}\tfrac{W}{L}(V_{GS}-V_{TH}) = \sqrt{2I_D\mu_n C'_{ox}\tfrac{W}{L}}$$

$$r_o = \tfrac{1}{g_D} = \tfrac{V_A}{I_D} \quad , \quad \lambda = \tfrac{1}{V_A} \quad , \quad \tfrac{g_m}{I_D} = \tfrac{2}{V_{GS}-V_{TH}} \quad , \quad C_{gs}^{lin} = \tfrac{1}{2}WLC'_{ox}$$

$$C_{gs}^{sat} = \tfrac{2}{3}WLC'_{ox} \, , \, C_{db} = \tfrac{C_{DB0}}{\sqrt{1+\frac{V_{DB}}{V_{bi}}}} \, , \, C_{sb} = \tfrac{C_{SB0}}{\sqrt{1+\frac{V_{SB}}{V_{bi}}}} \quad , \quad f_T = \tfrac{g_m}{2\pi(C_{gs}+C_{gd}+C_{gb})}$$

Optoelectrónica

$$E_{ph} = hf \quad , \quad c = \lambda f$$

Fotogeneración:

$$I = I_0\cdot e^{-\alpha x}$$

$$G_L(x,\lambda) = G_{L0}e^{-\alpha x}$$

$$\lambda[\mu m] = \tfrac{1.24}{E_g[eV]}$$

$$\tfrac{\partial\Delta p_n}{\partial t} = D_P\tfrac{\partial^2\Delta p_n}{\partial x^2} - \tfrac{\Delta p_n}{\tau_p} + G_L$$

$$n'(t) = n(0)\cdot e^{-t/\tau}$$

$$n'(x) = n(0)\cdot e^{-x/L_P}$$

Fotoconductores:

$$\Delta I = \Delta\sigma AE$$

$$\eta = G_L\cdot A\cdot L\cdot\tfrac{hf}{P_{opt}}$$

$$I_n = q\cdot\eta\cdot\tfrac{P_{opt}}{hf}$$

$$G_{opt} = \tfrac{\Delta I}{I_n} = \tfrac{\tau_n v_n}{L} = \tfrac{\tau_n\mu_n E}{L}$$

Fotodiodos:

$$I = I_{dark} + I_L$$

$$I = I_S\cdot\left(e^{V/V_T}-1\right) + I_L$$

$$I_L = -qAG_L\left(L_n + W + L_p\right)$$

$$\eta = \tfrac{I_L}{q}\cdot\tfrac{hf}{P_{opt}}$$

$$R = \tfrac{I_L}{P_{opt}} = \tfrac{q\eta}{hf}$$

Celdas solares:

$$P = I\cdot V$$

$$P = I_S\cdot V\left(e^{V/V_T}-1\right) - I_L\cdot V$$

$$FF = \tfrac{P_{max}}{I_{SC}\cdot V_{OC}} = \tfrac{V_m\cdot I_m}{I_{SC}\cdot V_{OC}}$$

$$\eta = \tfrac{P_m}{P_{inc}} = \tfrac{FF\cdot I_{SC}\cdot VOC}{P_{in}}$$